Daily Dough

HOA01: M-Commerce System & Architecture Setup

Academic System Architecture Specification, Data Communication Flow, and Tiered Design Breakdown

🏛️ **Institution:** University of Rizal System (URS)

📚 **Course:** ITE 7 – Application Development & Emerging Technologies

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📄 **Activity:** Hands-On Activity 01 (HOA01)

Project Overview & Executive Summary

Daily Dough is an on-demand mobile commerce platform specializing in artisanal handcrafted donuts and specialty espresso drinks. The platform employs a modern **Three-Tier M-Commerce Architecture** designed to provide seamless customer purchasing on Android mobile devices alongside complete managerial visibility and control for store administrators via a responsive web dashboard.

📘 Academic Architecture Note: Cloud Firestore vs. Realtime Database

While the baseline *ITE 7 Module* introduces Cloud Firestore, the Daily Dough platform implements **Firebase Realtime Database (RTDB)** via persistent REST/HTTP communication. This architectural choice delivers ultra-low-latency order lifecycle synchronization (Pending → Processing → Delivered) and enables the Single Page Web Dashboard to run efficiently with zero external library overhead.

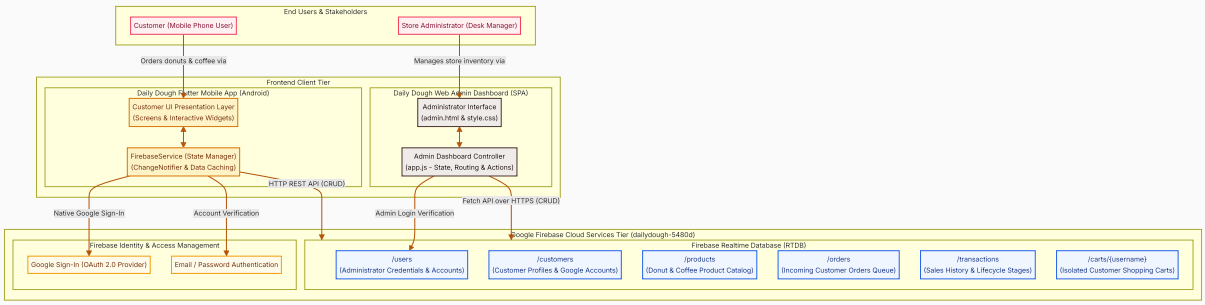
VISUAL REPRESENTATION LEGEND

- Human Actors & Roles (Pink)
- Mobile Flutter Client (Caramel/Amber)
- Web Admin Dashboard (Espresso Brown)
- Firebase Cloud Services (Gold)

- [ ] Realtime Database Endpoints (Blue)

Diagram A: Overall System Architecture

THREE-TIER ARCHITECTURE



Purpose & Overview

Provides a high-level view of the entire Daily Dough system. It illustrates the role separation between mobile customer shopping and web-based store management, connected through shared cloud services.

Key Components

- **Customer Mobile App:** Native Android client built with Flutter.
- **Web Admin Dashboard:** Desktop management SPA for store admins.
- **Cloud Backend (dailydough-5480d):** Firebase Authentication & Realtime Database.
- **Database Nodes:** Structured JSON endpoints for users, products, orders, and carts.

Data Flow

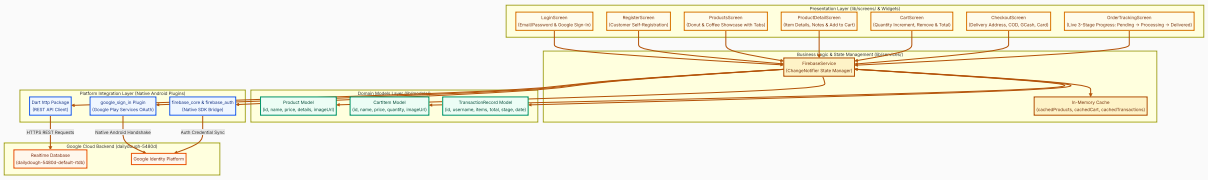
Customer requests flow from the Flutter app to the cloud database. The web admin dashboard queries the same endpoints to process orders and update menu availability.

HOA01 Curriculum Connection

Fulfills **HOA01 Objective 1** (Understanding M-Commerce structure) and matches the 3 major components described on **Page 3** of the ITE 7 module.

Diagram B: Mobile Application Architecture

FLUTTER CLIENT BREAKDOWN



Purpose & Overview

Illustrates the internal software architecture of the mobile shopping application, following clean Model-View-Service separation.

Key Components

- **Screens (`lib/screens/`):** 7 user-facing views handling the complete customer purchase lifecycle.
- **`FirebaseService`:** Central ChangeNotifier state manager with in-memory caching.
- **Domain Models:** Strongly-typed Dart classes for items, carts, and orders.
- **Plugins:** Platform channels linking native Android code to Dart.

Data Flow

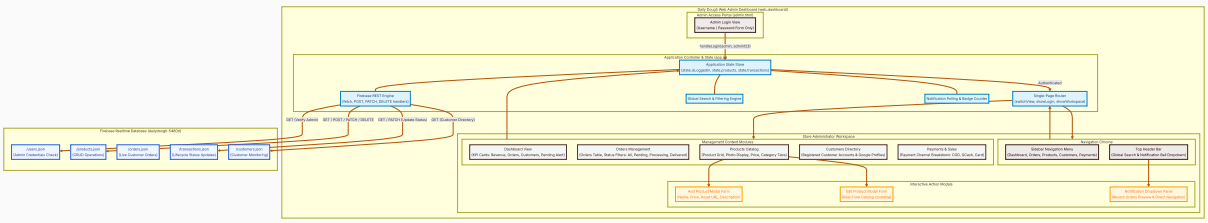
User gestures on screens invoke methods on `FirebaseService`. The service updates local cache, synchronizes with Firebase via HTTP REST, and triggers reactive UI updates.

HOA01 Curriculum Connection

Directly implements **HOA01 Activity 4 (Step 7: Create Folder Structure)**, organizing the codebase into `models`, `screens`, and `services`.

Diagram C: Web Admin Dashboard Architecture

MANAGERIAL SINGLE-PAGE APP



Purpose & Overview

Outlines the architecture of the Web Admin Dashboard, detailing how store staff manage inventory, review sales KPIs, and update customer order statuses.

Key Components

- **Admin Login:** Dedicated username/password portal protecting store data.
- **KPI Dashboard:** Live counters for sales, volume, customers, and pending alerts.
- **Catalog CRUD:** Add, edit, and delete donut/coffee listings in cloud database.
- **Orders Pipeline:** Interactive status workflow (Pending → Processing → Delivered).

Data Flow

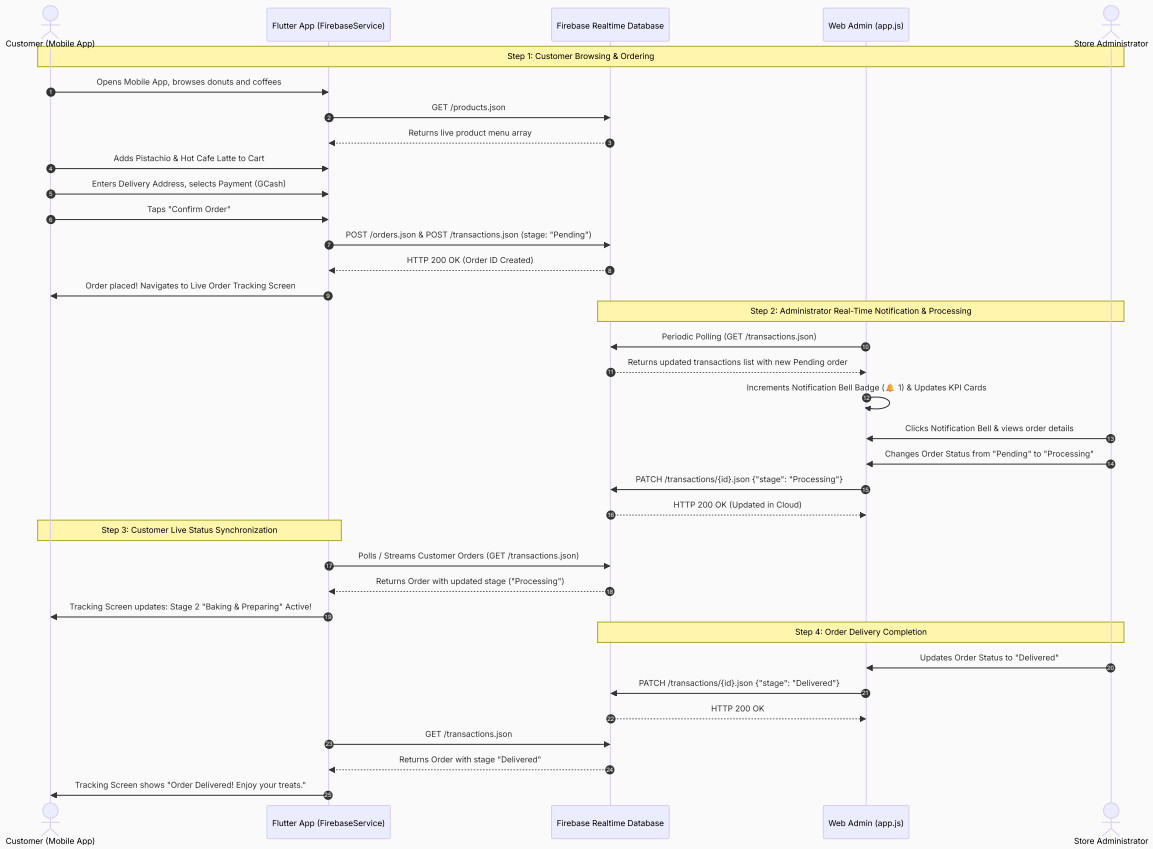
The single-page controller (`app.js`) handles client routing, manages session tokens, and executes REST calls directly against the cloud JSON database.

HOA01 Curriculum Connection

Fulfills the administrative monitoring and management requirements specified in **HOA01 Overview (Module Page 3)** and **HOA04 (Web-Based Admin Dashboard)**.

Diagram D: System Communication & Data Flow

END-TO-END ORDER LIFECYCLE



Purpose & Overview

Demonstrates the end-to-end communication sequence of an order from customer submission to final delivery.

Key Steps

- **1. Placing Order:** Mobile app writes order details to `/orders` and `/transactions`.
- **2. Admin Notification:** Dashboard detects new order, triggers bell badge and audio alert.
- **3. Kitchen Processing:** Admin updates status to *Processing*, advancing customer tracking UI.
- **4. Completion:** Admin marks order as *Delivered*, completing the transaction.

Data Flow

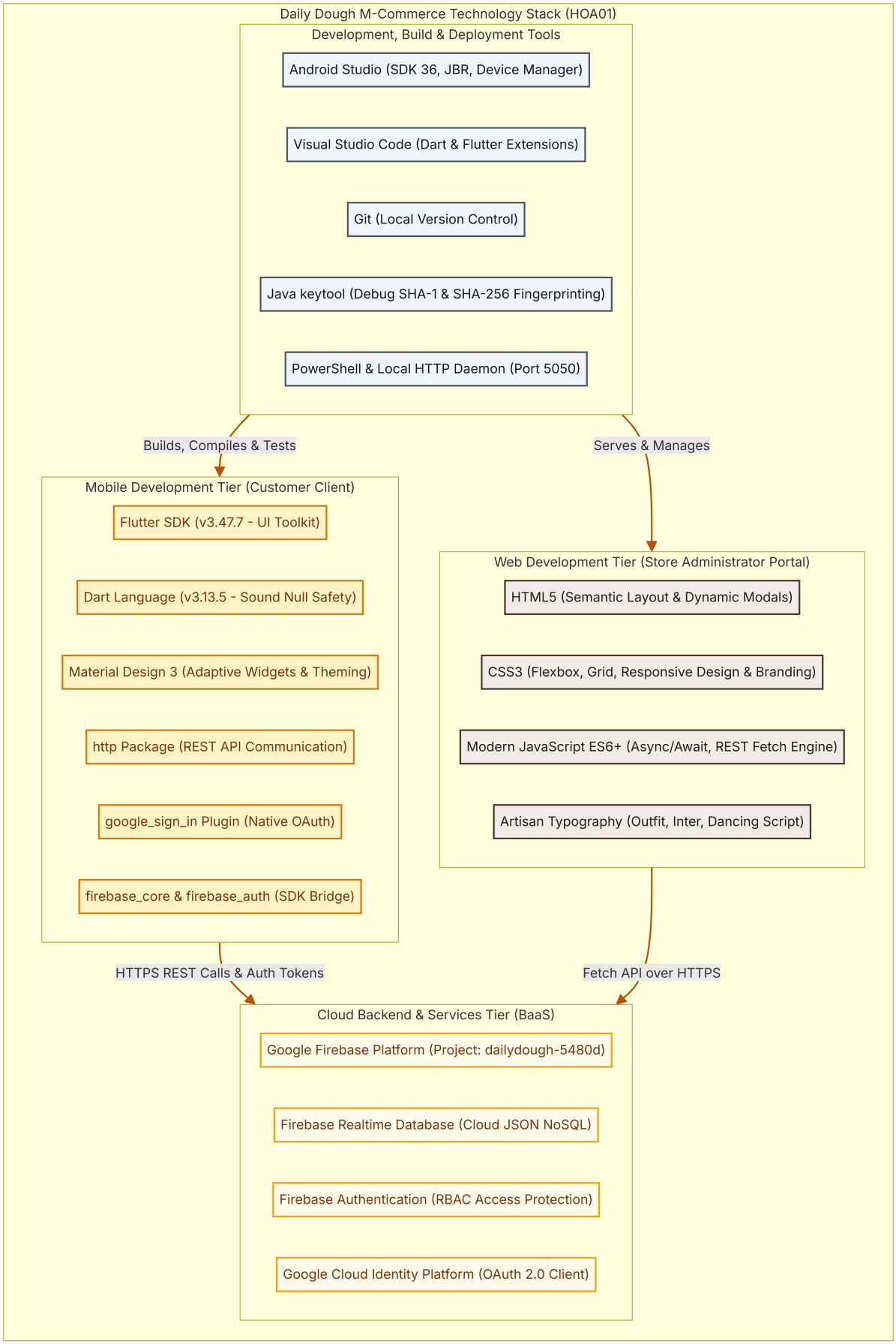
All state transitions occur asynchronously via cloud REST endpoints. No direct mobile-to-web connection is needed.

HOA01 Curriculum Connection

Demonstrates **Real-Time Order Synchronization** as defined in **HOA01 Outcomes (Page 8)** and **HOA05 (System Integration)**.

Diagram E: Technology Stack

ENGINEERING CLASSIFICATION


Daily Dough M-Commerce Technology Stack (HOA01)
Development, Build & Deployment Tools
Android Studio (SDK 36, JBR, Device Manager)
Visual Studio Code (Dart & Flutter Extensions)
Git (Local Version Control)
Java keytool (Debug SHA-1 & SHA-256 Fingerprinting)
PowerShell & Local HTTP Daemon (Port 5050)
Builds, Compiles & Tests
Serves & Manages
Mobile Development Tier (Customer Client)
Flutter SDK (v3.47.7 - UI Toolkit)
Dart Language (v3.13.5 - Sound Null Safety)
Material Design 3 (Adaptive Widgets & Theming)
http Package (REST API Communication)
google_sign_in Plugin (Native OAuth)
firebase_core & firebase_auth (SDK Bridge)
Web Development Tier (Store Administrator Portal)
HTML5 (Semantic Layout & Dynamic Modals)
CSS3 (Flexbox, Grid, Responsive Design & Branding)
Modern JavaScript ES6+ (Async/Await, REST Fetch Engine)
Artisan Typography (Outfit, Inter, Dancing Script)
HTTPS REST Calls & Auth Tokens
Fetch API over HTTPS
Cloud Backend & Services Tier (BaaS)
Google Firebase Platform (Project: dailydough-5480d)
Firebase Realtime Database (Cloud JSON NoSQL)
Firebase Authentication (RBAC Access Protection)
Google Cloud Identity Platform (OAuth 2.0 Client)

Purpose & Overview

Categorizes every programming language, runtime, cloud service, and developer utility used to build, test, and run Daily Dough.

Tier Categorization

- **Mobile Tier:** Flutter SDK, Dart, Material 3, Google Play Services.
- **Web Tier:** HTML5, CSS3, ES6+ JavaScript, custom artisan fonts.
- **Cloud Tier:** Google Firebase BaaS, Realtime Database, Firebase Auth.
- **Tooling:** Android Studio, Gradle Kotlin DSL, PowerShell, keytool.

Integration

Standardized HTTPS REST APIs and Google Play Services platform channels bridge frontend clients to cloud services.

HOA01 Curriculum Connection

Fulfills **HOA01 Activity 1 (Installing Required Software)** and **Activity 5 (Connecting Firebase)**.